

This documentation is structured as a **System Design Specification (SDS)**, formatted for integration into libraries of machine-readable maps.

Technical Specification: Multiplexed Entanglement Mesh (MEM)

1. Executive Summary

The MEM is a non-local, fault-tolerant transport layer designed for high-integrity data pipelines. It utilizes quantum state correlation to achieve zero-latency synchronization across geographically dispersed nodes, reinforced by a "Continuous Automatic Reinitialization" (CAR) protocol.

2. Structural Decomposition

The architecture is defined by a tiered hierarchy of sub-modules, managed by a centralized Master Orchestration Manifest.

2.1 The Connector Layer (Physical Interface)

- **Definition:** The transducer interface between local classical logic and the quantum state.
- **Constraint:** Must be treated as a **Singleton** per link to prevent pointer collision.
- **Key Interface:** Ghost_Pointer (A reference to a non-local StateVector).

2.2 The CAR Watchdog (Stability Layer)

- **Definition:** An asynchronous, non-blocking background process.
- **Behavior:** Monitors Correlation_Integrity metrics.
- **Event Trigger:** Upon decoherence (threshold < 0.95), the handler executes a Buffer-and-Swap operation, ensuring the application layer perceives no interruption in the data stream.

2.3 The Redundancy Fabric (Consensus Layer)

- **Definition:** A multiplexed array of entangled links managed by a Redundancy_Mesh_Manager.
- **Operational Logic:** Employs Weighted_State_Consensus. Data integrity is verified by comparing the outcomes of parallel links (Majority Vote/BFT).
- **Failover Policy:** Instantaneous weight reallocation upon link degradation.

3. Workflow Specification (MRP Library)

The following procedures are stored as callable modules in the library:

ID	Procedure	Function	Integrity Requirement
MRP_001	IntegrityPoll	Real-time Eigenstate stability check.	Absolute
MRP_002	PriorityMapping	SLA-based routing configuration.	High
MRP_003	LinkSelection	Deterministic path selection (N \ge 2).	Absolute
MRP_004	SyncDispatch	Atomic state write across Mesh.	Critical
MRP_005	Verification	Post-transmission parity audit.	Critical

4. Master Orchestrator (Schema)

This is the primary form used to initiate any transmission. It acts as the "Controller" for the procedures above.

```
{
  "Manifest": "SIS_MEM_Orchestrator_v1",
  "Execution_Pipeline": [
    "MRP_001_IntegrityPoll",
    "MRP_002_PriorityMapping",
    "MRP_003_LinkSelection",
    "MRP_004_SyncDispatch",
    "MRP_005_Verification"
  ],
  "Global_State_Management": {
    "Buffer_Path": "/system/sis/buffer/active_state",
    "Logging_Destination": "/var/log/sis/forensic_audit_trail"
  },
  "Error_Handling": {
    "Strategy": "Autonomous_CAR_Rehydration",
    "Max_Retries": 3,
    "Escalation_Path": "System_Admin_Notify_High_Alert"
  }
}
```

5. Implementation Standards

- **Unit Testing:** Every procedure (MRP) must be tested against a "Decoherence Injection" scenario, where link failure is simulated to ensure the CAR Handler initiates a clean hand-off.
- **Storage:** All manifests must be stored in an immutable ledger format to preserve the forensic chain of custody for every transaction performed across the mesh.

Integration Directive

As you deploy this, remember that the **AARM** framework relies on these nodes as the edges of your relationship graph. When the Verification step (MRP_005) flags a mismatch, the resulting audit entry should be tagged with the Link_ID that caused the decoherence, allowing the SIS to dynamically blacklist unstable paths in the geospatial map.